

Task 1: EDA

01. Data Overview

The dataset contains **1,20,759 rows and 22 original columns** (the 23rd column, `_year`, is a redundant copy of `Year`), covering reported crime incidents from **2021-01-01 to 2025-12-31**. Each row is a single reported incident, uniquely identified by `ID` (no duplicates) and `Case Number` (a small number of cases share a number).

Field	Type	Description
ID / Case Number	int / text	Unique record and police case identifiers
Date	datetime (text)	Date & time of incident, format MM/DD/YYYY HH:MM:SS AM/PM
Block	text	Anonymized street block
IUCR / FBI Code	categorical code	Illinois / FBI standardized offense classification codes
Primary Type / Description	categorical	High-level crime category and detailed sub-type
Location Description	categorical	Setting of the incident (street, apartment, store, etc.)
Arrest / Domestic	boolean	Whether an arrest was made / whether it was domestic-related
Beat / District / Ward / Community Area	numeric codes	Nested Chicago policing & administrative geographies
X/Y Coordinate, Latitude/Longitude, Location	numeric / text	Spatial position
Year / Updated On	int / datetime	Incident year + last record-updatetimestamp

02. Missing Value Analysis

Field	Missing Rows	
X Coordinate / Y Coordinate / Latitude / Longitude / Location	1877	
Location Description	463	
Community Area	11	

03. Data Quality Analysis

- **No duplicate records :**

Zero fully duplicated rows and zero duplicated **ID** values.

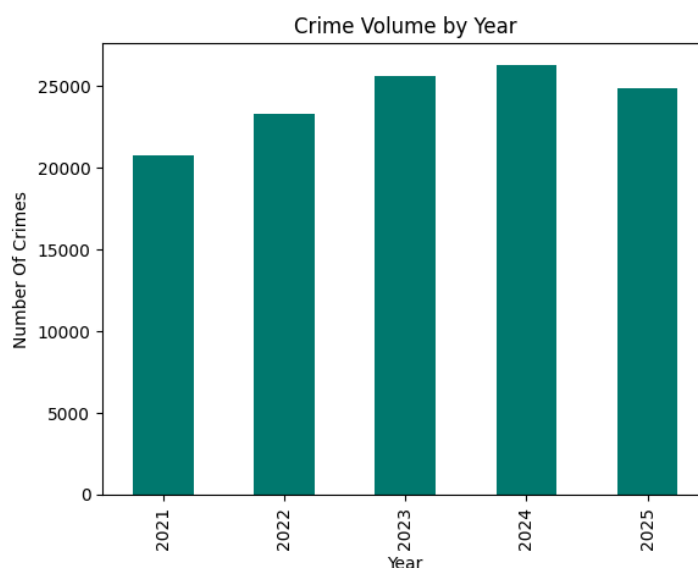
Four **Case Number** values appear twice : these are all *HOMICIDE* records where a single case includes two victims logged as separate incidents. This is expected police record-keeping behavior, not a data error.

- **Geographic codes are internally consistent:**

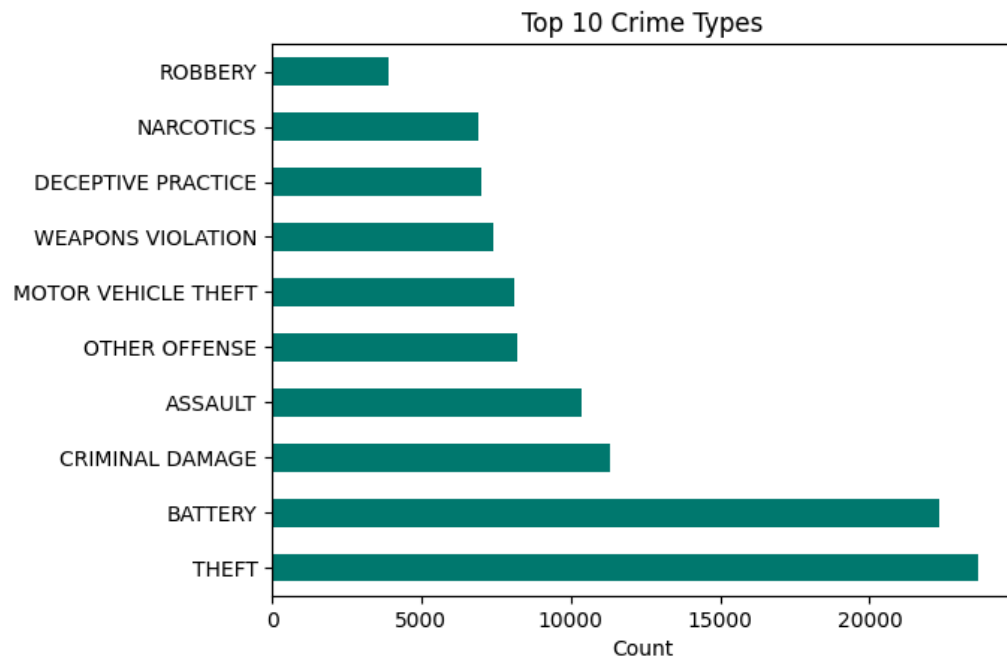
District (1–31), Ward (1–50), Community Area (1–77) and Beat all fall within Chicago's official administrative ranges, with no negative, zero, or out-of-range codes.

04. Data Distribution Analysis

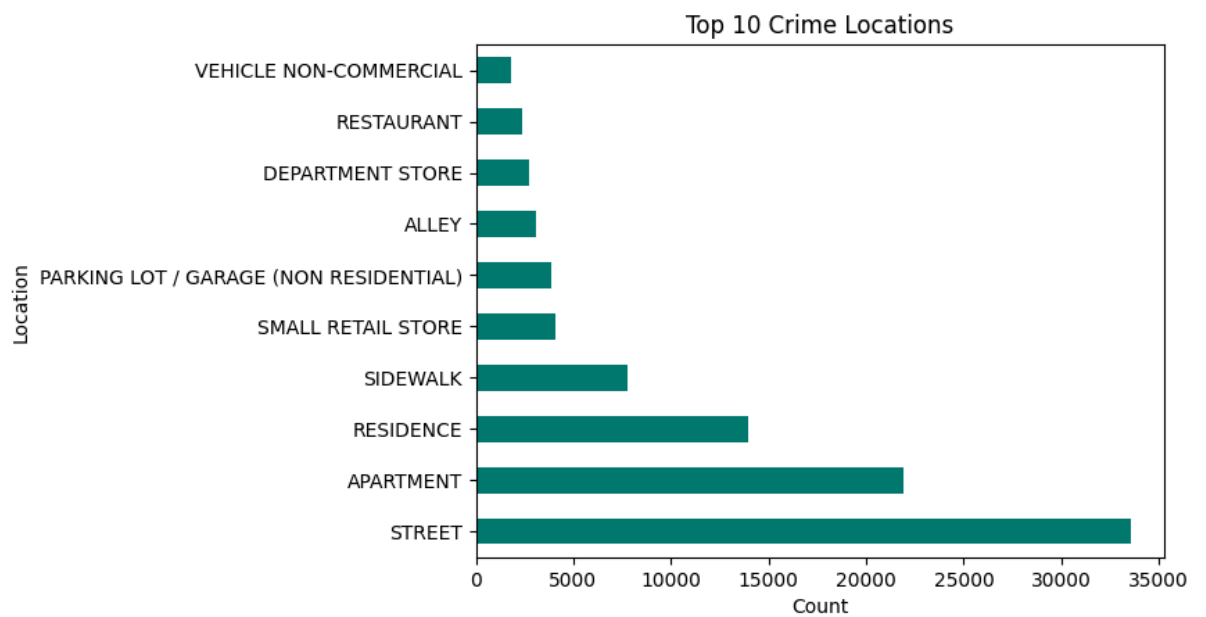
1. **Crime Volume by Year**



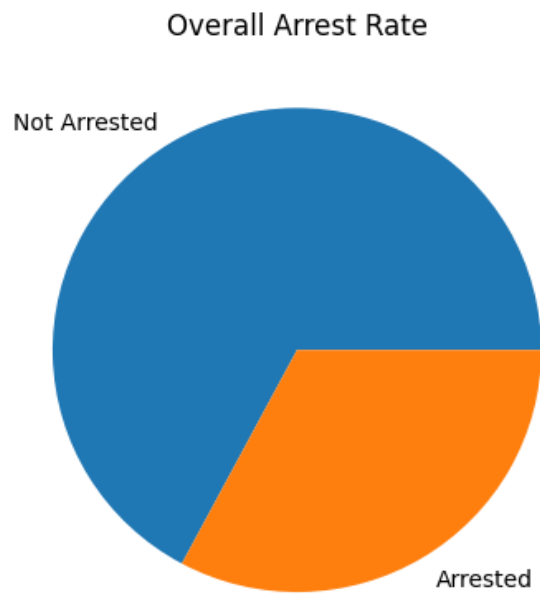
2. Crime Type Distribution:



3. Location Distribution:



4. Overall Arrest Rate:



05. Univariate Analysis